

Analysis 2: HW6

Aamod Varma

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Problem 3. We have u is continuous and that,

$$\int_{R^n} u \Delta \psi = 0$$

we need to show that u is smooth and $\Delta u = 0$.

We can approximate u using the convolution between u and ϕ where we define $\phi = \frac{1}{\varepsilon^n} f$ where $\int f = 1$. And now note that we get $u_\varepsilon = u * \phi$ but if we take n to infinity then we have the approximation of $u_\varepsilon \rightarrow u$. Now note that,

$$\Delta u_\varepsilon = u * \Delta \phi = \int_{R^n} u(y) \Delta \phi(x - y) dy$$

but the RHS is zero so we have $\Delta u_\varepsilon = 0$. But now use the notating from (1) to get,

$$u_\varepsilon(x) = \frac{1}{|B(x, r)|} \int_{B(x, r)} u_\varepsilon(y) dy$$

and take $\varepsilon \rightarrow 0$ then as we have $u_\varepsilon \rightarrow 0$ we have u also gets the same property.

So we showed that u is smooth and $\Delta u = 0$